

# Practicum 1: Perceptron Learning and Decision Boundary Dynamics

## Objective

Implement the Perceptron Learning Algorithm (PLA) from scratch to classify standard 2-input Boolean logic gates. You will manually document the mathematical weight updates and visually track the geometric rotation of the decision boundary hyperplane as the network processes linearly separable and inseparable datasets.

## The Weight Update Rule

You must apply the explicit error-driven update rule, incorporating the learning rate  $\alpha$  and the input features  $x_{ij}$ . For this assignment, set the learning rate to  $\alpha = 1$  to easily observe discrete boundary shifts.

- **Weight Update:**  $w_j := w_j + \alpha(y - \hat{y})x_{ij}$
- **Bias Update:**  $b := b + \alpha(y - \hat{y})$

(Note:  $\hat{y}$  is the predicted output using the Heaviside step function where  $\hat{y} = 1$  if  $z \geq 0$ , and  $y$  is the target ground truth. The update only occurs when  $y \neq \hat{y}$ .)

## Assignment Tasks

### Task 1: The AND Gate

- Initialize parameters.
- Execute the PLA for the 2-input AND truth table.
- Continue looping through epochs until the stopping criterion of convergence (zero errors in a full epoch) is met.

### Task 2: The OR Gate

- Initialize parameters.
- Execute the PLA for the 2-input OR truth table.
- Run until perfect convergence is achieved.

### Task 3: The XOR Gate

- Initialize parameters.
- Execute the PLA for the 2-input XOR truth table.
- Because XOR is linearly inseparable, enforce a hard stopping criterion of exactly **10 full epochs** to document the oscillation of the parameters.

## Task 4: The Half Adder

A Half Adder adds two single binary digits, outputting a **Sum** and a **Carry**.

- Realize the **Carry** bit using one perceptron.
- Realize the **Sum** bit using another perceptron.
- Train the two independent perceptrons simultaneously for a maximum of **10 epochs**. Document which perceptron converges and which fails.

## Required Deliverables for All Tasks

### 1. Step-by-Step Execution Tables

Construct a tabular trace for *every single row* of the truth table across every epoch. You must explicitly list:

- Current Input Vector ( $x_1, x_2$ )
- Target Label ( $y$ )
- Net Input Calculation ( $z = w_1x_1 + w_2x_2 + b$ )
- Model Prediction ( $\hat{y}$ )
- The applied update calculation and the resulting new parameters ( $w_1, w_2, b$ ).

### 2. Geometric Boundary Plots

You must generate a 2D Cartesian plot ( $x_1$  vs.  $x_2$ ) for **each step** where a weight update actually occurs. Every plot must clearly display:

- The four data points correctly plotted and color-coded by their target class (Class 0 vs. Class 1).
- The current weight vector  $W$ .
- The decision boundary hyperplane.

## Submission Guidelines

- **Document Format:** Compile all deliverables (mathematical execution tables and geometric boundary plots) into a single, well-organized **PDF document**.
- **File Naming Convention:** Name your final submission file strictly as `RollXX_FirstName_Assign1.pdf` where XX is your class roll number; e.g. `Roll01_Adam_Assign1.pdf` for a person named Adam Smith having class roll 01.